

Nicholas Wise Saragih Sumbayak

Bandung, Indonesia | nicholasaragih@gmail.com | +62 812 82799631 | [linkedin.com/in/nicholaswises/](https://www.linkedin.com/in/nicholaswises/) | nicwise.dev

Informatics Engineering Sophomore at Institut Teknologi Bandung exploring **scalable software infrastructure**, event-driven architectures, and **distributed systems**. Demonstrated **technical leadership** in engineering high-volume e-commerce platforms and robust transaction backends. Proven track record of leading development teams and optimizing full-stack performance using modern tech stacks like **Go**, **TypeScript**, and **React**.

EXPERIENCE

Aku Masuk ITB 2026

Deputy Head of IT

Bandung, Indonesia

Oct 2025 – Feb 2026

- Coordinated a **12-person team** to develop a comprehensive **e-commerce platform** for **event registration** and **merchandise sales**.
- Reduced admin operational workload by **~55%** by building an **internal dashboard** with **CSV order exports**, **attendance QR scanning**, **stock monitoring**, and **role-based access controls**, replacing a spreadsheet-and-WhatsApp workflow with a single self-service interface.
- Engineered the core transaction backend to handle high-volume spikes of **7,000+ concurrent users**, maintaining **99.95% uptime** during peak ticket drops.
- Engineered the transaction backend on Hono and Drizzle ORM with PostgreSQL, achieving **98.7% cart-to-payment conversion**, raising over **Rp120M+** in total sales.
- Implemented containerized automated deployments to a VPS via Docker Compose, **reducing manual overhead by ~70%** and slashing staging-to-production deployment times from **30 minutes to under 4 minutes**.
- Tech: **React**, **Vite**, **Bun**, **TypeScript**, **Hono**, **PostgreSQL**, **DrizzleORM**

TEDxITB 9.0

Director of Website/IT

Bandung, Indonesia

Sep 2025 – May 2026

- Architected a capacity-gated ticketing backend with tiered pricing (Early Bird vs Regular) and automated bundle availability checks, **eliminating 100% of oversell incidents** and sustaining a **98% successful checkout rate** across 4 ticket release windows.
- Boosted event publicity reach by 50%** through comprehensive **SEO optimization**, reducing Largest Contentful Paint (LCP) by **40%** to hit a **93/100 Lighthouse performance score**.
- Optimized operational workflows by designing a scalable ticketing backend, implementing tRPC query batching that **reduced database connection pool contention by 35%**.
- Automated post-purchase communications via Resend API with React Email templates, **reducing manual attendee follow-up work by ~85%** and ensuring **100% delivery** of payment confirmations, event reminders, and QR ticket emails within 60 seconds of order completion.
- Architected a robust checkout system that successfully processed **100+ orders**, reducing user checkout drop-off rate by **18%** and amassing a total of **Rp20M+** from ticket and merchandise purchases.
- Tech: **T3 Stack (Next.js, TypeScript, TailwindCSS, tRPC)**, **PostgreSQL**, **DrizzleORM**

PROJECTS

Mjolnir - Digital Web-based Library Platform

- Engineered a zero-framework Go HTTP backend and React/TypeScript frontend, implementing custom Singular Value Decomposition (SVD) from scratch to eliminate external math dependencies and **reduce memory footprint by ~60%**.
- Built LSA text and PCA image recommendation engines projecting features into a **50-component latent space**, enabling accurate semantic and visual cosine similarity lookups in under **12ms**.
- Eliminated 90-second model cold-starts** via gob binary caching and containerized the full stack with multi-stage Docker builds for streamlined, single-command deployments.

DOM Vector - DOM Tree Visualizer

- Designed a Go/Gin and React Flow SSR application deployed on Azure, featuring a custom recursive-descent CSS selector parser that evaluates complex compound queries in **under 3ms**.
- Optimized tree operations with parallelized BFS/DFS traversals (yielding a **2.8x speedup**) and binary lifting for Lowest Common Ancestor (LCA) queries, **slashing latency** from **~45ms** to **<1.5ms** on **5,000+ node trees**.
- Automated deployment with multi-stage **Docker Compose** and **Caddy reverse proxy** providing automatic HTTPS certificate provisioning, enabling one-command deployments to Azure VMs via CI shell scripts.

EDUCATION

Institut Teknologi Bandung

B.Eng in Computer Science · GPA 3.69 / 4.00

Aug 2024 – Jul 2028 (Expected)

- Relevant Coursework** - Object-Oriented Programming, Operating Systems, Software Engineering, Algorithm Strategy

TECHNICAL SKILLS

Core Languages	TypeScript, Go, C++, Python, Java, Haskell
Frontend Development	React, Next.js, TailwindCSS, React Query, Zod, Zustand, Motion
Backend & Databases	Express.js, Hono, Gin, FastAPI, Flask, PrismaORM, DrizzleORM, MongoDB, PostgreSQL, MySQL, Supabase, tRPC, Redis
Infrastructure, Cloud & Tools	Docker, Kubernetes, Azure, Cloudflare R2, Linux, Git, GitHub, Figma, Postman